

A new modelling framework for fruit and nut tree phenology

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Abstract

Many fruit and nut tree species depend on low temperatures during the cold season to overcome dormancy. Lacking chill may result in diminished bud burst rates and uncoordinated blooming, both of which can cause severe yield losses. With global warming leading to rising winter temperatures in many growing regions, assessments of temperature requirements during dormancy and projections of tree phenology have gained renewed attention. A number of studies in recent decades have attempted to improve the predictive performance of phenology models and to integrate eco-physiological as well as genetic processes. Nevertheless, the development of modeling frameworks that are readily applicable to data are still lacking. As a result, many practitioners must use outdated models, which have not been parameterized for specific species and/or varieties, and have also not been validated for the climate of interest. To help close the gap between scientific developments and the need for applied analyses, we recently re-implemented the Dynamic Model of chill accumulation in the widely distributed programming environment R. We linked this sub-model with a heat model to project bloom dates and developed a framework for automated model fitting. Here, we present the application of the framework to long-term data sets of bloom dates of sweet cherry (*Prunus avium*) to demonstrate the use of the model and its predictive performance.

Keywords: dormancy, blooming, dynamic model, growing degree hours, chilling requirements, *Prunus avium*

INTRODUCTION

Many temperate fruit and nut trees evolved in regions that are characterized by cold winters. To avoid the freezing of their leaves and flowers, trees fall dormant during the cold season (Campoy et al., 2011). For dormancy to be broken, species- and variety-specific chill (Vegis, 1964) and heat (Anderson et al., 1986) requirements have to be fulfilled. If these needs are not met, production is usually not economically viable.

In recent years, evidence has been mounting that many tree species are severely affected by climate change (Luedeling et al., 2011; Parmesan and Yohe, 2003; Benmoussa et al., 2018; Lorite et al., 2020). With given prospects for climate warming, it seems unlikely that the current array of tree cultivars grown in many of the world's orchards will allow economically viable production over the coming decades (Benmoussa et al., 2018). Winter warming has thus emerged as an important concern for management of fruit and nut trees in many regions (Luedeling, 2012; El Yaacoubi et al., 2014). Consequently, assessments of temperature requirements during dormancy and projections of tree phenology are of increasing relevance for orchard management.

A number of recent studies have attempted to improve the quality of phenology models and conducted comparative studies (e.g. Darbyshire et al., 2017; Gaeta et al., 2018; Kwon et al., 2020; Fernandez et al., 2020b). However, the development of transparent and easily applicable modelling frameworks has failed to keep up with this elevated process understanding (Egea et al., 2020). Consequently, practitioners often end up using simple and

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outdated models. These are rarely parameterized for the species and/or varieties of interest (Lorite et al., 2020), and are seldom validated in the climate of interest. These unvalidated (Anderson et al., 1986) and mis-parameterized models have severely restricted the accuracy of previous attempts to model tree responses to temperature, especially in warm growing regions (Erez, 2000; Fadón et al., 2020).

Here, we provide a synthesis of the state of the art on these issues and some possible solutions. We propose an integrated phenology projection model that builds on the structure of the most advanced chill (Fishman et al., 1987) and heat models (Luedeling and Brown, 2011) and allows for rapid and context specific parameterization and validation.

MATERIALS AND METHODS

Modelling framework

The model is an equation-based, dynamic model that predicts bloom dates in response to hourly temperature data. It is implemented in the programming language R (R Core Team, 2019) to allow for easy handling. The model consists of three sub-models: (i) a chill model, (ii) a heat model, and (iii) a link-model for the former two.

The chill model closely follows the 'Dynamic model' developed by Fishman et al. (1987). In short, it describes the accumulation of chill as a two-step process. The first being a reversible process of formation and destruction of some thermally labile virtual substance x described by the following differential equation:

$$\frac{d}{dt} x(t) \equiv x(t) = k_0(t) - k_1(t) \times (t) \quad (1)$$

Here, $x(t)$ is the labile substance in units of its critical value and t the time. The equation describes the change in the amount of x , where the first term, $k_0(t)$ represents the amount of x that is formed and the second term, $-k_1(t)x(t)$, represents the amount of x that is destroyed, which depends on both k_1 and the actual amount of x at time t . k_0 and k_1 are temperature-dependent and species-specific rate constants for formation and destruction, respectively, and are assumed to conform to the Arrhenius law:

$$k_i = A_i \cdot e^{-E_i/T(t)} \quad i=0,1 \quad (2)$$

with $T(t)$ the time-dependent temperature in Kelvin, A_i time-independent amplitudes and E_i the time-independent activation energies measured in Kelvin. In the second step of the accumulation of chill, once a certain amount of x is formed, a fraction of this amount is converted to a quantum of a stable substance y . The size of the fraction is temperature-dependent and follows a sigmoidal response curve, defined by the transition temperature T_f and the slope s . For endo-dormancy to be broken, a species- and cultivar-specific amount of the stable substance y has to be accumulated. For details, we refer to Fishman et al. (1987).

Once endo-dormancy is broken, a certain amount z_c of heat needs to be accumulated during eco-dormancy for the trees to flower, usually described by a heat model. Here, we use the Growing Degree Hours model (GDH) developed by Anderson et al. (1986). According to this model, there is a minimum temperature T_b , a maximum temperature T_c and an optimal temperature T_u that define how effectively heat is accumulated at a given temperature T (see Figure 1a). The total amount of accumulated heat is denoted as z and the weight function P_z is defined as:

$$P_z(T) = \begin{cases} \frac{1}{2} \left(1 + \cos \left(\pi + \pi \frac{T-T_b}{T_u-T_b} \right) \right), & T_b < T \leq T_u \\ 1 + \cos \left(\frac{\pi}{2} + \frac{\pi}{2} \frac{T-T_u}{T_c-T_u} \right), & T_u < T \leq T_c \\ 0, & \text{otherwise} \end{cases} \quad (3)$$

For linking the chill and heat models, we assume a gradual transition from endo- to eco-dormancy. The more chill portions have been accumulated by the plant, the more effectively the heat hours can be accumulated following a sigmoidal response (see Figure 1b):

$$P_y(y) = \frac{s_y}{1+s_y}, s_y = \exp\left(s_1 y_c \frac{y-y_c}{y}\right) \quad (4)$$

Here, y_c is the critical value for y and s_1 is the slope at $y=y_c$. Bloom takes place once a critical amount z_c is accumulated.

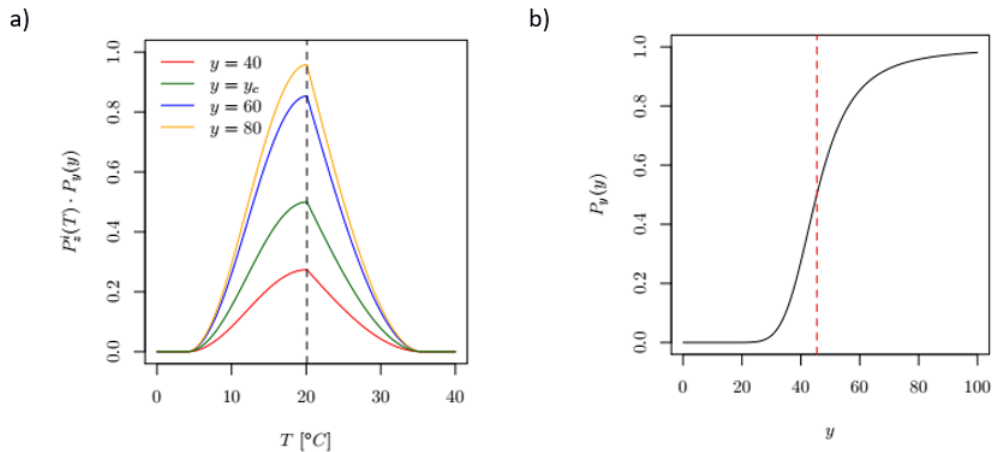


Figure 1. a) The accumulation of heat $P_z(T) * P_y(y)$ in response to temperature T for different values of y following the GDH model. The dashed line denotes the optimal temperature T_u . b) Sigmoidal function that describes the effectiveness of heat accumulation $P_y(y)$ depending on the accumulated amount of chill y . The dashed red line denotes y_c , the critical value of y for which the slope s_1 is parameterized.

This model approach includes 12 model parameters that can be fitted to species- and variety-specific responses to temperature regimes (Table 1).

Table 1. Model parameters with short descriptions and their starting values for parameterization and fitted values.

Parameter	Description	Starting values	Fitted value
E_0 (K)	Activation energy of formation of the labile substance x following the Arrhenius law	3372.8	3850.2
E_1 (K)	Activation energy of destruction of x	9900.3	10462.0
A_0 (1/h)	Amplitude of formation of x	6319.5	9927.4
A_1 (1/h)	Amplitude of destruction of x	$5.9399 \cdot 10^{+13}$	$5.9398 \cdot 10^{+13}$
T_f (°C)	Transition temperature defining the transition from labile substance x to stable substance y	4.0	3.94
s (1/K)	Slope of sigmoidal response of transition	1.6	23
y_c	Critical value for stable substance y indicating inflection point of heat accumulation efficiency	40.0	37.1
s_1 (1/K)	Slope at $y=y_c$ defining the gradual shift from chill to heat accumulation	0.5	3.2
T_u (°C)	Optimal temperature for heat accumulation	25.0	16.9
T_b (°C)	Minimum temperature for heat accumulation	4.0	5.7
T_c (°C)	Maximum temperature for heat accumulation	36.0	18.9
z_c (h)	Critical amount of heat for blooming	190.48	242.31

Data

Flowering and temperature data were recorded from 1984 to 2018 at Campus Klein-Altendorf, a research station of Bonn University (6°59' E, 50°37' N, 150 m a.s.l.). This area lies in the south of the Lower Rhine Bay; the climate is warm-temperate. For this study, we use data from the sweet cherry (*Prunus avium* subsp. *duracina*) cultivar 'Schneiders Späte Knorpelkirsche'. This cultivar is a late-flowering cherry with high economic importance in Germany. The flowering data were recorded yearly. The frequency of observations within years is unknown. Full bloom was determined on the basis of the BBCH-scale (Blanke and Kunz, 2009) and defined as level 65 when 50% of flower buds were blooming.

Air temperature was measured at 2 m height in a weather station on the campus, following the measurement protocols of the German Meteorological Service. Gaps in the temperature data were interpolated and the records were converted to an hourly time step using the 'stack_hourly_temps' function from the 'chillR' R-package (Luedeling, 2018). Figure 2 shows average, minimum and maximum temperatures for the observation site in January, the coldest month.

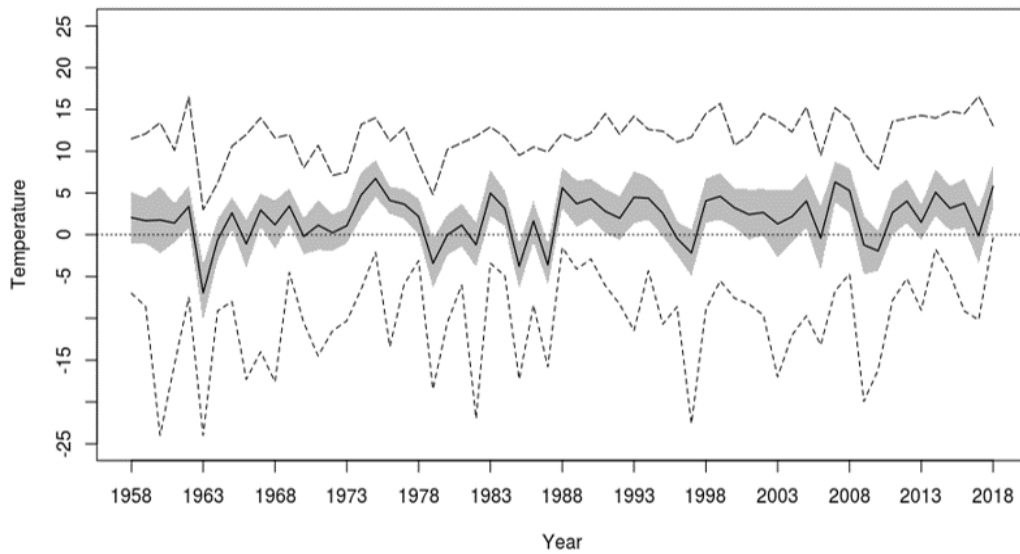


Figure 2. Temperature in January across observation years. Solid line represents mean temperature in January, dashed lines absolute maximum and minimum temperatures and gray area the range between average daily minimum and average daily maximum temperatures within months.

Parameterization and validation

The data set was split into a training set containing 24 data points (1984-2008, 1985 missing) and a validation set with 10 data points (2009-2018). We used the training set to estimate the 12 parameter values of the model. This was done by minimizing the residual sum of squares (RSS) using a generalized simulated annealing algorithm implemented in the 'GenSA' R-package (Xiang et al., 2013). RSS is defined as:

$$RSS = \sum (\Delta)^2 = \sum (\text{Day}_{\text{obs}} - \text{Day}_{\text{pred}})^2 \quad (5)$$

Using the fitted parameters, we then projected the bloom days for the years 2009 to 2018 based on the temperature data and compared these projections with the observational data.

RESULTS AND DISCUSSION

Overall, we could demonstrate that it is possible to parameterize our phenology model reasonably well to the observed bloom dates from 1984 to 2008, even though the relation of the number of parameters to the number of data points is rather large. The best fit we obtain has $RSS=262$ with the average deviation between model and data amounting to 2.33 days. While the RSS -value is relatively large, the qualitative structure of the observed data, i.e. the way in which trees respond to temperature regimes, is well reproduced by the model (Figure 3a). This qualitative match is confirmed by the strong correlation between model prediction and observation data (Pearson's $r=0.89$) (Figure 3b).

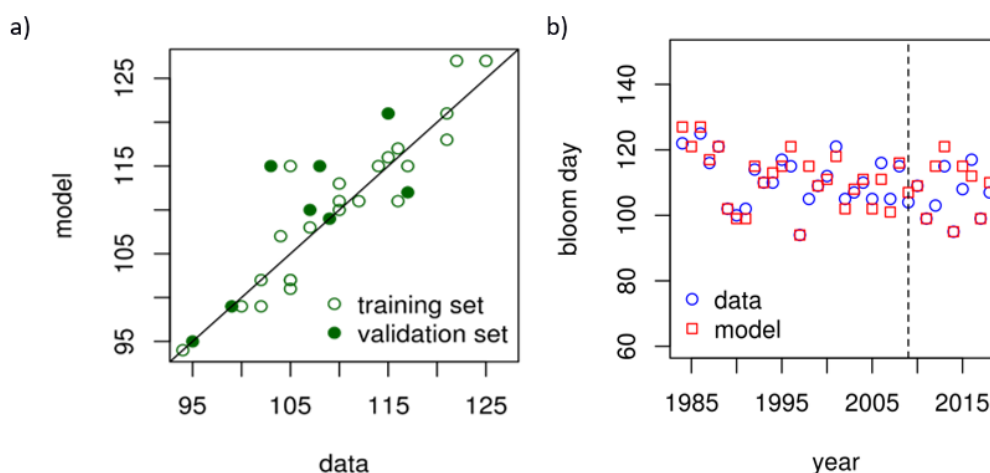


Figure 3. a) Bloom date (Julian day) as a function of year. Blue circles represent the observed blooming days, red squares indicate the model prediction. The vertical dashed line marks the last year included in the parameterization of the model. b) Observed bloom days versus predicted bloom days. Open symbols indicate data included in the fit, filled symbols represent validation data.

The average deviation between projected and observed bloom dates for the years 2009 to 2018 was 3.6 days, being slightly larger than the average deviation in the training data. Consequently, the correlation between data and model prediction for the years 2009 to 2018 was also slightly lower, but with a correlation coefficient of Pearson's $r=0.82$. This is still reasonably good, especially given the fact that we extrapolate to years with warmer average temperatures.

When interpreting the results, apart from uncertainty in the model structure and parameterization, another source of error that should be considered is error in the observation data. Unfortunately, for our data, we do not know the frequency of observations of the trees within years.

It may well be that trees were not checked every day, which would lead to a lack of precision in the data. In this case, neither an improvement in the model structure itself nor in the parameterization procedure could compensate for the potential error in the response data.

When comparing the fitted parameter values to the initial guesses (Table 1), two aspects attract attention. First, both, the optimal and the maximum temperature for heat accumulation, T_u and T_c , respectively, are quite low and the efficiency of heat accumulation drops quickly above the optimal temperature (Figure 4a). While these low values warrant further studies on a tree's heat response during dormancy, they provide an indication that the heat response function by Anderson et al. (1986), which – to the best of our knowledge – was developed for non-dormant trees, is a poor description of how trees perceive heat during this period. Second, the parameters of the link-model connecting chill- and heat-accumulation indicate a very steep transition from endo- to eco-dormancy, almost resembling a step-

function (Figure 4b). The flexibility of the modeling framework allowed us to find a response here known as the sequential model (Ashcroft et al., 1977) as the best fitting model, even though all other gradations between sequential and parallel chill and heat accumulation would also have been allowed by the model structure. We note, however, that the climate at Klein-Altdorf, with abundant chill accumulation and a long periods of low temperatures beyond what is necessary for most trees to fulfill their chilling requirements, may not have presented many situations where trade-offs between chill and heat would have manifested.

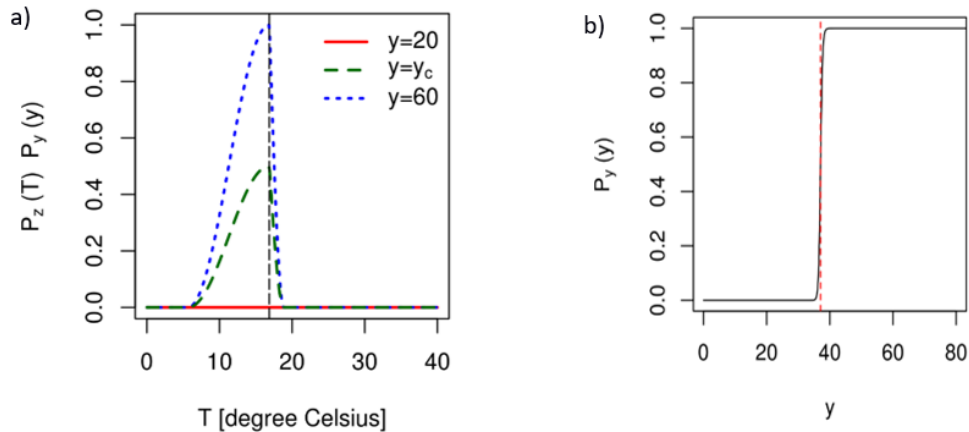


Figure 4. a) Heat accumulation efficiency as a function of temperature and the amount of accumulated chill portions y according to the link-model and the Growing Degree Hours model. The dashed line indicates the optimal temperature T_u . b) The weight function $P_y(y)$ shows the transition between endo- and eco-dormancy as a function of y , the amount of accumulated chill units. The dashed line indicates the critical value y_c .

While both the model validation and identified parameter values indicate a solid parameterization of the model, in the scope of this study, we did not check the robustness of the parameterization procedure and the potential covariance of parameters. There are several possibilities for increasing the quality of results. Longer time series comprising a wide range of temperature regimes may ensure more stable and reliable fits. Another approach would be to bundle data from different cultivars and assume some of the parameters to be constant across them. These parameters can then be fitted with the pooled cultivar-specific observations and then be used to produce more stable values for the remaining parameters. Finally, assessing the timing of the end of endo-dormancy, e.g. through forcing experiments, would strongly restrict the number of potential parameter combinations leading to good fits and hence enable more effective parameterization.

This study presents one of the first applications of a comprehensive bloom prediction framework that combines the state of the art in chill and heat modelling. It allows for a gradual transition between the chilling and forcing phases. This has been indicated experimentally (Fernandez et al., 2020a; Harrington et al., 2010; Pope et al., 2014) but is not yet implemented in the temperature models commonly used. The major innovation of this approach is that all parameter values can be derived from data, including the interaction between chill and heat, thus allowing comprehensive deduction of all important temperature response parameters from observed temperature and bloom data.

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